

CAMPUSLOOP AI

An AI-Enhanced Trusted Campus Second-Hand Marketplace

PROJECT PROPOSAL

Introduction to Software Engineering Assessment

Group Number	5
Programme of Study	BSc (AI)
Team Size	10 students
Proposal Date	27 September 2026
Estimated Project Period	28 September 2026 - 1 November 2026

Group Member	Student ID	Group Member	Student ID
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This document describes planned work. It does not claim that any proposed feature, integration or evaluation result has already been completed.

1 Project Title and Problem

Project title: CampusLoop AI - An AI-Enhanced Trusted Campus Second-Hand Marketplace

CampusLoop AI is a proposed web platform for trusted second-hand trading within a university community. It combines a conventional marketplace and a traceable transaction process with explainable assistance for search, matching and price reference. The basic marketplace remains usable when optional AI services are unavailable.

Problem

Students currently buy and sell textbooks, electronics, furniture and daily necessities through fragmented channels such as messaging groups, social feeds and general marketplaces. Listings use inconsistent descriptions, become outdated quickly and are difficult to search. Buyers may describe a purpose and budget while sellers describe a brand or model, so keyword-only discovery can miss relevant items.

After discovering an item, students still need to negotiate, agree a meeting time and place, confirm the handover and resolve unsuccessful interactions. Informal channels provide limited support for structured offers, transaction status, two-party confirmation, reviews, contextual reports or auditable administrator decisions. This creates unnecessary coordination cost and weakens trust.

The project is important because a campus marketplace can reduce student living costs, extend the useful life of goods and support a local circular economy. However, these benefits require clear permissions, privacy protection, transaction consistency and honest AI limitations rather than a visually impressive but unreliable demonstration.

The proposed course scope excludes real payment, delivery, university identity-provider integration and automatic punitive decisions. Campus identity is simulated for demonstration, and any risk signal is routed to human review.

2 Proposed Solution

CampusLoop AI will provide one browser-based workflow for public discovery, authenticated trading and administrator governance. Guests may browse public marketplace and wanted information. Registration or login is required only for actions that depend on identity, including publishing, editing, favouriting, chatting, negotiating, ordering, reviewing and reporting.

Core solution components

- Account and permission layer: registration, login, profile, student and administrator roles, session expiry and field-level privacy.
- Marketplace and wanted requests: listings, images, categories, condition, price, search, filters, favourites, seller management, expiry and lifecycle states.
- Transaction coordination: contextual chat, structured offers and counter-offers, order creation, meetup versions, two-party completion, reviews and notifications.
- Governance: contextual reporting, evidence visibility, administrator review, reasoned decisions, appeals and audit fields.
- Explainable assistance: keyword and semantic retrieval, wanted-to-product matching, price-reference ranges, transparent factors and deterministic non-AI fallback.
- Engineering foundation: React and TypeScript frontend; FastAPI backend; PostgreSQL and optional pgvector; generated OpenAPI SDK; Redis, object storage and workers where justified; CI, logs, backup and recovery instructions.

The backend and database will remain the sources of truth for identity, permission, listing availability, accepted offer, order state, meetup version, completion confirmation and review eligibility. Frontend state and AI suggestions are derived views and cannot override authoritative business facts.

The system will be developed as a modular proof of concept. Optional services must have visible unavailable states and safe fallbacks so that browsing and the basic transaction loop do not fail solely because an intelligent function, cache or worker is unavailable.

3 Objectives

ID	Specific and achievable objective
O1	Define and implement account, profile, role, session and privacy boundaries with explicit authorised and unauthorised results.
O2	Provide product and wanted-request publishing, browsing, keyword search, filtering, favourites and lifecycle management through a responsive interface.
O3	Demonstrate a real two-account transaction from chat and structured negotiation through order creation, meetup agreement, independent completion confirmation and review.
O4	Integrate at least two justified intelligent functions with a simple baseline, versioned input and output, explanation, evaluation measure and unavailable-service fallback.
O5	Provide contextual reporting and administrator review while preventing automated risk signals from directly changing accounts, orders or penalties.
O6	Deliver a reproducible engineering package containing OpenAPI, generated SDK, migrations, seed data, tests, CI evidence, clean-deployment instructions, backup and recovery guidance, technical documentation and a final demonstration.

Acceptance basis

Objectives will be accepted only from an integrated repository commit. A feature author performs initial checks, another group member reviews the pull request, M10 independently executes the relevant acceptance scenarios and M1 confirms that the result remains within the agreed scope. Mock data may support Week 2 prototypes but cannot close a Week 3 real-integration objective.

The primary demonstration will use a buyer and seller account to publish, discover, discuss, negotiate, confirm a meetup, complete the exchange, review one another and create a contextual report. Permission, invalid-state, duplicate-action and service-unavailable scenarios will also be shown.

4 Expected Benefits

Stakeholder	Expected benefit
Student buyers	Faster discovery, persistent wanted requests, clearer price references, structured negotiation and safer coordination.
Student sellers	Standardised listings, access to relevant buyers, controlled listing states, traceable offers and evidence from completed transactions.
Campus community	Lower reuse costs, reduced waste, clearer behaviour expectations and a practical local circular-economy channel.
Administrators	Contextual reports, controlled evidence access, auditable review decisions and clear separation between allegations, automated signals and human outcomes.
Project team	Experience across requirements, architecture, frontend, backend, databases, AI evaluation, testing, deployment, documentation and team integration.

Expected indicators of success

- A non-author can start the documented environment and apply migrations without private instructions from the original developer.
- Two accounts can complete the agreed MVP transaction using real APIs and persistent data.
- Non-participants cannot access private conversations, orders or report evidence.
- Accepted AI functions have reproducible baseline comparisons, explanations and safe fallback behaviour.
- The final release has no unresolved P0 or P1 defect and includes regression, load, recovery and known-limitation evidence.

The project will not claim commercial readiness, guaranteed fraud detection, guaranteed price accuracy or unrestricted production capacity. Benefits will be reported only to the extent supported by executed evidence.

5 Timeline

The project uses five consecutive one-week phases from 28 September 2026. The dates are estimates and the official MyAberdeen deadline takes precedence. Each gate requires a reviewable increment, evidence and handoff; incomplete work keeps an owner and recovery action.

Phase	Estimated dates	Main activities	Expected completion
Phase 1 / Week 1	28 Sep - 4 Oct	Confirm scope, roles, journeys, domain and AI feasibility, risks and tests.	Approved Phase 1 baseline.
Phase 2 / Week 2	5 - 11 Oct	Create clickable prototypes; freeze API, SDK, database, environment, AI and test contracts.	Runnable contract baseline.
Phase 3 / Week 3	12 - 18 Oct	Implement authentication, marketplace and the real two-account transaction flow.	Persisted MVP loop.
Phase 4 / Week 4	19 - 25 Oct	Integrate two justified AI functions; test metrics, explanation, privacy and fallback; freeze.	Frozen release candidate.
Phase 5 / Week 5	26 Oct - 1 Nov	Run regression, security, load, recovery and clean deployment; finish documents.	Tagged submission package.

6 Action Plan

The action plan assigns a primary owner to each work area. Progress is shown as Planned because this proposal describes the agreed work programme and does not claim that a phase has already been completed.

Responsible member	Required activity	Deadline	Status
M1 - Huang Qiwei	Requirements, scope, traceability, integration gates, risks, final report and presentation.	4 Oct; weekly; 1 Nov	Planned
M2 - Chen Zihong	Frontend shell, navigation, auth/profile, shared components and page states.	11 Oct; 18 Oct	Planned
M3 - Gao Junjie	Marketplace, listings, wanted requests, search, matching and price-advice UI.	11 Oct; 18 Oct; 25 Oct	Planned
M4 - Hu Keming	Chat, offers, orders, meetup, completion, reviews, reports and notifications UI.	11 Oct; 18 Oct	Planned
M5 - Li Mingyuan	Authentication, sessions, privacy, roles, field visibility and administrator permissions.	11 Oct; 18 Oct	Planned

Cross-group dependency order

Rules and contracts are published before consumers integrate them: M5/M6 provide identity and business contracts; M7/M8 provide versioned intelligent-function outputs; M9 provides the environment and migrations; M2-M4 integrate the user experience; M10 independently tests the integrated result; M1 controls scope and the phase handoff.

6 Action Plan - continued

Responsible member	Required activity	Deadline	Status
M6 - Ma Xuan	Marketplace/transaction entities, APIs, WebSocket, states, transactions and idempotency.	11 Oct; 18 Oct; 25 Oct	Planned
M7 - Chen Hongqing	Keyword/semantic retrieval, wanted matching, explanations and search evaluation.	11 Oct; 18 Oct; 25 Oct	Planned
M8 - Luo Kangrui	Price reference, trust evidence, risk-review assistance, evaluation and fallback.	11 Oct; 18 Oct; 25 Oct	Planned
M9 - Liang Jiawen	Database, migrations, seed data, environment, storage, jobs, CI, deployment and recovery.	11 Oct; 18 Oct; 1 Nov	Planned
M10 - Ye Zidan	Independent QA, defects, permission/E2E tests, load/recovery tests and release sign-off.	Weekly; 1 Nov	Planned

Academic integrity and AI use

This proposal represents Group 5's planned work. Generative AI assisted with structure, wording, formatting and consistency checks. The group will review and approve the final submission and remains responsible for its accuracy and originality. Any external text, code, figure, image, data or idea used during the project will be acknowledged according to University guidance. This proposal contains no external figure or unverified result.

Supervisor approval

The group will discuss the scope, five-week schedule and technical approach with the assigned supervisor before submission. Any substantial later change will also be discussed and recorded.